Temporary proof appendix for the transverse and fixed map analyses

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DO NOT INCLUDE IN THE MANUSCRIPT.
This appendix is a separate proof audit. It does not modify the claims or the numerical tables in `paper.tex`.

1 Scope

This appendix audits two coefficient statements used in the proposed PXL analysis. The first statement concerns the degree two Level I shape. It shows that a transverse affine restriction has full affine coefficient support. The second statement concerns the two degree four Level I shapes. It gives an exact criterion for full homogeneous coefficient support on a fixed slice. The criterion fails after only six variables are fixed.

These are algebraic coefficient statements. They do not say that the official coefficient distribution is uniform. They also do not prove a solving degree, a Macaulay rank prediction, or a PXL running time. We record these limits again after each proof.

2 The degree two affine coefficient map

Let $q = 19$ and let $A = \mathbb{F}_{q^2}$. We regard every A space as an $\mathbb{F}_q$ space when taking symmetric forms. Put

$$V = A^v, \qquad h = v - 2.$$

Let U be the full common column domain and let $\pi_v : U \to V$ be the public vinegar projection. Choose an $\mathbb{F}_q$ subspace $L \subseteq U$ of dimension h, with basis $\ell_1, \ldots, \ell_h$, and define its extension span by

$$L_A = \mathrm{Span}_A\{\pi_v(\ell_1), \ldots, \pi_v(\ell_h)\} \subseteq V.$$

Definition 2.1 (Extension linear transversality)**.** The direction L is extension linearly independent if

$$\dim_A L_A = h.$$

For such a direction, an affine coset $a + L$ is transverse if

$$\pi_v(a) \notin L_A.$$

The first condition is stronger than injectivity of $\pi_v|_L$ over $\mathbb{F}_q$. It says that the A lines generated by the vinegar projections are independent. The second condition adds the A line generated by the affine offset.

2.1 Uniform coset incidence

Lemma 2.2 (Exact transverse incidence). *Fix an extension linearly independent direction L of dimension h. If $a + L$ is a uniform affine coset of L in U, then*

$$\Pr[\pi_v(a) \in L_A] = |A|^{h-v}.$$

For $h = v - 2$, the failure probability is 19^{-4}.

Proof. A uniform point of U induces the uniform distribution on U/L, since every coset has the same number of representatives. The event $\pi_v(a) \in L_A$ is independent of the chosen representative because $\pi_v(L) \subseteq L_A$. The induced vinegar coset is therefore uniform in V/L_A. Exactly one of the $|A|^{v-h}$ cosets is L_A itself. Hence

$$\Pr[\pi_v(a) \in L_A] = \frac{|L_A|}{|V|} = |A|^{h-v}.$$

Substituting $|A| = 19^2$ and $v - h = 2$ gives 19^{-4}. $\square$

This calculation samples a uniform coset first and rejects a nontransverse coset. It does not replace the root sampling experiment by a distribution conditioned on transversality.

2.2 A blockwise support lemma

We first isolate the linear algebra used by the coefficient map. For an $\mathbb{F}_q$ bilinear form $C : A \times A \to \mathbb{F}_q$, write

$$C^{\mathrm{sym}}(\xi, \eta) = C(\xi, \eta) + C(\eta, \xi).$$

Lemma 2.3 (The mixed block). *The map*

$$\mathrm{Hom}_{\mathbb{F}_q}(A \otimes_{\mathbb{F}_q} A, \mathbb{F}_q) \longrightarrow \mathrm{Sym}^2_{\mathbb{F}_q}(A)^*, \qquad C \longmapsto C^{\mathrm{sym}},$$

is surjective. Its kernel is the one dimensional space of alternating bilinear forms on A.

Proof. The characteristic is odd, so a symmetric form S has preimage $S/2$. The kernel consists exactly of forms satisfying $C(\xi, \eta) = -C(\eta, \xi)$. These are the alternating forms. Since A has dimension two over $\mathbb{F}_q$, their space has dimension $\binom{2}{2} = 1$. $\square$

Proposition 2.4 (Full affine support for one degree two base form). *Assume that L is extension linearly independent and that $a+L$ is transverse. The restriction map from the fresh native symmetric vinegar coefficients of one base form to its three descended channel polynomials is surjective onto*

$$\mathbb{F}_q[x_1, \ldots, x_h]^3_{\le 2}.$$

Equivalently, every monomial of degree at most two can be assigned an arbitrary coefficient in $\mathrm{Sym}^2_{\mathbb{F}_q}(A)^*$.

Proof. Put $b = \pi_v(a)$ and $e_i = \pi_v(\ell_i)$. Transversality gives the direct sum of A lines

$$Ab \oplus Ae_1 \oplus \cdots \oplus Ae_h \subseteq V.$$

Let B be one fresh native symmetric $\mathbb{F}_q$ bilinear form on this sum. For $\xi, \eta \in A$, evaluate the corresponding channel on

$$b + \sum_{i=1}^{h} x_i e_i.$$

The channel label is symmetric in (ξ, η), so its coefficient belongs to $\mathrm{Sym}^2_{\mathbb{F}_q}(A)^*$. The independent blocks of B supply the coefficients as follows.

The diagonal block on Ae_i is an arbitrary symmetric form on the two dimensional $\mathbb{F}_q$ space A. It supplies the coefficient of x_i^2. The block between Ae_i and Ae_j is an arbitrary bilinear form on $A \times A$. Its contribution to $x_i x_j$ is its symmetrization. Theorem 2.3 shows that this contribution covers all three channel coefficients. Its only unused direction is the one dimensional alternating kernel.

The block between Ab and Ae_i is governed by the same mixed block map. It therefore supplies an arbitrary coefficient of x_i in all three channels. The diagonal block on Ab supplies the three constant coefficients. All listed blocks are independent because the displayed sum of A lines is direct. We may set every unused block to zero and extend B arbitrarily to the rest of V. This constructs a preimage for every triple of affine quadratic polynomials. □

Corollary 2.5 (The Level I degree two support statement)**.** *For $(v, o, \ell, r) = (48, 16, 2, 2)$, take $h = 46$. A uniform coset fails transversality with probability 19^{-4}. Conditional on transversality, each independent base form maps onto three arbitrary affine quadratics in 46 variables. The sixteen base forms therefore map onto the coefficient space of 48 affine quadratics in 46 variables.*

Proof. The probability is Theorem 2.2. The native coefficient blocks for distinct base forms are independent parameters. Apply Theorem 2.4 to each form and take their direct sum. □

Descent to the official coefficient space. The block map in Theorem 2.4 is defined over $\mathbb{F}_q$. One may also verify its surjectivity after extending scalars to a splitting field. If C is its cokernel, scalar extension gives $C \otimes_{\mathbb{F}_q} \Omega = 0$. The extension $\Omega/\mathbb{F}_q$ is faithfully flat, so $C = 0$. Thus the split channel calculation descends to the official $\mathbb{F}_q$ coefficient constraints.

Distributional limit. Surjectivity gives full support when the native coefficient source has full support. It gives uniform output only when the native source is uniform on the relevant finite vector space. The official byte reduction source is not uniform on $\mathbb{F}_{19}$. This proposition does not bound its distance from a uniform MQ distribution.

3 The degree four fixed map

Put $k = \mathbb{F}_{19}$ and let Ω be a splitting field for $A = \mathbb{F}_{19^4}$. We first work over Ω. The four split eigenblocks are spaces V_a, with planted oil spaces $O_a \subseteq V_a$, for $0 \le a < 4$. For both Level I degree four shapes,

$$\dim(V_a/O_a) = v = 28.$$

The total block dimensions are 33 for $(28, 5, 4, 4)$ and 32 for $(28, 4, 4, 5)$.

Let X be the space of main variables on a fixed slice. A realization of the slice gives four linear maps

$$T_a : X \longrightarrow V_a.$$

Write $q_a : V_a \to V_a/O_a$ for the quotient map and put

$$\overline{T}_a = q_a \circ T_a.$$

For one native UOV base form, its split coefficient space is the direct sum of the diagonal blocks

$$\mathcal{B}_{aa} = \{B \in \mathrm{Sym}^2(V_a^*) : B|_{O_a \times O_a} = 0\}$$

and the cross blocks

$$\mathcal{B}_{ab} = \{C \in \mathrm{Hom}(V_a \otimes V_b, \Omega) : C|_{O_a \times O_b} = 0\}, \qquad a < b.$$

Put

$$\mathcal{B} = \bigoplus_a \mathcal{B}_{aa} \oplus \bigoplus_{a<b} \mathcal{B}_{ab}.$$

Equivalently, put $V = \bigoplus_a V_a$ and $O = \bigoplus_a O_a$. Then $\mathcal{B}$ is the space of symmetric forms on V that vanish on $O \times O$, written in its four diagonal and six cross blocks. The fixed realization $T = (T_0, T_1, T_2, T_3)$ defines the linear coefficient map

$$\Phi_T : \mathcal{B} \longrightarrow \bigoplus_{0 \le a \le b < 4} \mathrm{Sym}^2(X^*).$$

On a diagonal block it is pullback by T_a. On a cross block its value is the symmetric form

$$(x, y) \longmapsto C(T_a x, T_b y) + C(T_a y, T_b x).$$

The target has four diagonal channels and six cross channels.

3.1 The fixed map criterion

Lemma 3.1 (One diagonal block)**.** *Let $T : X \to V$ and let $O \subseteq V$. The pullback map*

$$\{B \in \mathrm{Sym}^2(V^*) : B|_{O \times O} = 0\} \longrightarrow \mathrm{Sym}^2(X^*)$$

is onto if and only if the composite $X \to V/O$ is injective.

Proof. Suppose that a nonzero $x \in X$ maps into O. Every form in the image then vanishes at (x, x). This is a proper linear condition on $\mathrm{Sym}^2(X^*)$, so the pullback map is not onto.

Conversely, suppose that $X \to V/O$ is injective. Choose a complement $V = O \oplus C$. The projection of $T(X)$ to C is an isomorphic copy of X. Given $S \in \mathrm{Sym}^2(X^*)$, define a symmetric form on this copy by transporting S. Extend it to C, and set its blocks involving O to zero. The resulting form vanishes on $O \times O$ and pulls back to S. □

Lemma 3.2 (One cross block)**.** *Let $T_a : X \to V_a$ and $T_b : X \to V_b$. If both maps $X \to V_a/O_a$ and $X \to V_b/O_b$ are injective, then the cross block map in* (3) *is onto* $\mathrm{Sym}^2(X^*)$.

Proof. Choose complements $V_a = O_a \oplus C_a$ and $V_b = O_b \oplus C_b$. Let $c_a : X \to C_a$ and $c_b : X \to C_b$ be the projections of T_a and T_b. Both maps are injective. Given $S \in \mathrm{Sym}^2(X^*)$, define a bilinear form on $c_a(X) \times c_b(X)$ by

$$C(c_a x, c_b y) = \frac{1}{2} S(x, y).$$

Extend this form to $C_a \times C_b$, and set every block involving an oil space to zero. The extension lies in $\mathcal{B}_{ab}$. Its symmetric pullback is S because the characteristic is odd. □

Theorem 3.3 (Fixed map linear surjectivity)**.** *The map Φ_T in* (3) *is onto if and only if every quotient map*

$$\overline{T}_a : X \longrightarrow V_a/O_a$$

is injective.

Proof. The source and target of Φ_T are direct sums over the ten channel blocks. If every $\overline{T}_a$ is injective, the four diagonal blocks are onto by Theorem 3.1. The six cross blocks are onto by Theorem 3.2. The blocks use independent source coordinates, so their direct sum is onto.

For the converse, suppose that some $\overline{T}_a$ is not injective. The projection of Φ_T to diagonal channel (a, a) is not onto by Theorem 3.1. Hence Φ_T is not onto. □

For five independent native base forms, the full map to the 50 channel quadrics is the direct sum of five copies of Φ_T. It is onto under the same four injectivity conditions.

Corollary 3.4 (The six variable fixing cannot give full support)**.** *For both Level I degree four shapes, fixing six variables in a* 37 *dimensional slice leaves* $\dim X = 31$*. The fixed coefficient map* Φ_T *is not onto for any realization* T*.*

Proof. Every quotient V_a/O_a has dimension 28. No map from a 31 dimensional space to a 28 dimensional space is injective. Apply Theorem 3.3. □

Thus six fixed variables do not give full homogeneous coefficient support. This criterion allows full support only after the main variable space has dimension at most 28. Starting from dimension 37, this requires fixing at least nine variables and choosing a realization that satisfies the four injectivity conditions.

3.2 The open good locus and realization

Proposition 3.5 (Open nonempty locus)**.** *Let* $n = \dim X$ *and let*

$$\mathcal{P} = \prod_{a=0}^{3} \mathrm{Hom}(X, V_a)$$

be the space of abstract four block realizations. The locus on which Φ_T *is onto is Zariski open. It is nonempty in* $\mathcal{P}$ *if and only if* $n \leq v$*.*

If the SNOVA construction supplies only a realization locus $\mathcal{R} \subseteq \mathcal{P}$*, the good locus in that family is* $\mathcal{R} \cap \mathcal{P}^{\mathrm{good}}$*. It is open in* $\mathcal{R}$*. Its nonemptiness requires one realizable tuple with all four quotient maps injective.*

Proof. By Theorem 3.3, the good locus is the intersection of the four full rank loci for $q_a T_a$. Each full rank condition is Zariski open. If $n > v$, every one of these loci is empty. If $n \leq v$, choose an injection $X \to V_a/O_a$ for each a and lift it through q_a. This gives a point in the good locus of $\mathcal{P}$.

The last statement follows by intersecting with $\mathcal{R}$. Nonemptiness in the ambient product does not show that the constrained realization family meets the good locus. □

3.3 Descent to the base field

Proposition 3.6 (Faithfully flat descent for a realized map)**.** *Suppose there are finite dimensional* k *spaces* $\mathcal{B}_k$ *and* $\mathcal{C}_k$*, and a* k *linear map*

$$\Phi_{T,k} : \mathcal{B}_k \longrightarrow \mathcal{C}_k,$$

whose scalar extension to Ω *identifies with the assembled split map* Φ_T*. Then* $\Phi_{T,k}$ *is onto if and only if* Φ_T *is onto.*

Proof. Let C be the cokernel of $\Phi_{T,k}$. Scalar extension is exact, so the cokernel of Φ_T is $C \otimes_k \Omega$. If Φ_T is onto, this tensor product is zero. The extension Ω/k is faithfully flat, so $C = 0$. The converse follows from right exactness of scalar extension. $\square$

The qualification in Theorem 3.6 is necessary. The individual eigenblocks and oil spaces can exist only after passing to the splitting field. One must first show that the assembled ten channel map is the scalar extension of a map defined over k. An abstract good tuple over Ω does not produce a k rational slice, and faithful flatness does not prove that the official byte reduction source is uniform.

3.4 Varying slice incidence calculations

The rest of this section records the earlier codimension and tangent calculations. These calculations allow the slice intersections to vary with the target quadrics. They do not prove that the coefficient map for a fixed slice is onto, and they do not change Theorem 3.4.

Let Z be the 37 dimensional slice before the six PXL coefficient variables are fixed. Write

$$Z = E \oplus X, \qquad \dim E = 6, \qquad \dim X = 31.$$

The four Frobenius eigenprojections are maps

$$\rho_a : Z \longrightarrow V_a, \qquad 0 \le a < 4.$$

For the shape $(28, 5, 4, 4)$, a generic ρ_a has a four dimensional kernel. For the shape $(28, 4, 4, 5)$, it has a five dimensional kernel.

Lemma 3.7 (Diagonal channel image on the varying slice)**.** *Let $\rho : Z \to V$ be surjective with kernel K. Pullback identifies $\mathrm{Sym}^2(V^*)$ with the symmetric forms S on Z for which $K \subseteq \operatorname{rad} S$. If $\dim Z = 37$ and $\dim K = c$, its codimension in $\mathrm{Sym}^2(Z^*)$ is*

$$\binom{38}{2} - \binom{38-c}{2}.$$

This codimension is 142 for $c = 4$ and 175 for $c = 5$.

Proof. A pulled back form vanishes whenever either input lies in K. Conversely, such a form descends uniquely to $Z/K \cong V$. The dimension is therefore $\binom{38-c}{2}$. Subtraction from $\dim \mathrm{Sym}^2(Z^*) = \binom{38}{2}$ gives the formula and the two stated values. $\square$

Lemma 3.8 (Alternating cross lift on the varying slice)**.** *Let $K_a, K_b \subseteq Z$ be disjoint subspaces of dimension c. A symmetric form S on Z is induced by a bilinear block that descends through*

$$(Z/K_a) \times (Z/K_b)$$

if and only if

$$S|_{K_a \times K_a} = 0, \qquad S|_{K_b \times K_b} = 0.$$

The cross channel image has codimension

$$2\binom{c+1}{2} = c(c+1).$$

This codimension is 20 for $c = 4$ and 30 for $c = 5$.

Proof. The two vanishing conditions are necessary. For sufficiency, choose an alternating form A_0 whose values on $K_a \times Z$ and $Z \times K_b$ cancel the corresponding values of S. The prescriptions agree on $K_a \times K_b$. They are compatible with alternation on K_a and K_b because the two displayed restrictions of S vanish. Since $K_a \cap K_b = 0$, extend this partial alternating form to all of Z.

The bilinear form $C = S + A_0$ vanishes on $K_a \times Z$ and on $Z \times K_b$. It therefore descends through the two quotients. Its quadratic polynomial is unchanged because $A_0(x, x) = 0$. The restriction maps to $\mathrm{Sym}^2(K_a^*)$ and $\mathrm{Sym}^2(K_b^*)$ are independent for disjoint subspaces. Their total dimension gives the codimension. □

There are four diagonal channels and six cross channels for each base form. The preceding lemmas give the following codimension audit.

Table 1: Codimension of the split coefficient family on the 37 dimensional varying slice.

Shape	Diagonal	Cross	Per base form	Five base forms
$(28, 5, 4, 4)$	142	20	$4(142) + 6(20) = 688$	3440
$(28, 4, 4, 5)$	175	30	$4(175) + 6(30) = 880$	4400

Lemma 3.9 (Injectivity before the oil quotient)**.** *There is a nonempty Zariski open set of decompositions $Z = E \oplus X$ with $\dim X = 31$ for which*

$$X \cap \ker \rho_a = 0 \qquad \textit{for every } a.$$

This holds for both Level I degree four shapes.

Proof. For one kernel of dimension c, the condition $X \cap K = 0$ is open on $\mathrm{Gr}(31, Z)$. It is nonempty when $31 + c \leq 37$. Here c is four or five. Each such open set is dense because the Grassmannian is irreducible. Their intersection is therefore nonempty. On this open set, every $\rho_a|_X$ is injective. □

Let $O_a \subseteq V_a$ have dimension five for $(28, 5, 4, 4)$ and four for $(28, 4, 4, 5)$. For a generic injective image $\rho_a(X)$, its intersection with O_a has dimension

$$\dim(\rho_a(X) \cap O_a) = 31 + \dim O_a - \dim V_a = 3.$$

Its pullback is

$$I_a = \ker(X \xrightarrow{\rho_a} V_a \to V_a/O_a).$$

Thus injectivity before the oil quotient is compatible with a three dimensional kernel after that quotient. Every one of the five diagonal base forms vanishes on the fixed space I_a.

The next incidence calculation permits this three space to vary with the five target quadrics.

Proposition 3.10 (Five quadrics with a varying common three space)**.** *Let X have dimension* 31 *over an algebraically closed field of odd characteristic. The projection*

$$\{(I, Q_1, \ldots, Q_5) : I \in \mathrm{Gr}(3, X),\ Q_s|_{I \times I} = 0\} \longrightarrow \mathrm{Sym}^2(X^*)^5$$

is dominant.

Proof. The Grassmannian has dimension

$$\dim \mathrm{Gr}(3,31) = 3(31-3) = 84.$$

For a fixed I, the five isotropy conditions impose $5\binom{4}{2} = 30$ linear equations. This count leaves room for dominance, but it does not prove it. We now exhibit a point where the projection has surjective differential.

Choose a three space I. In X/I, choose five pairwise disjoint three spaces $W_1, \dots, W_5$. Their total dimension is 15, while $\dim(X/I) = 28$. For each s, choose Q_s so that its pairing between I and W_s is perfect. Set its pairing between I and W_t to zero when $t \neq s$, and impose $Q_s|_{I\times I} = 0$.

A tangent vector to the Grassmannian at I is a map $\delta : I \to X/I$. The derivative of the sth isotropy equation is

$$(e, e') \longmapsto Q_s(\delta e, e') + Q_s(e, \delta e').$$

The component of δ in W_s can be chosen independently of its components in the other W_t. The perfect pairing identifies $\mathrm{Hom}(I, W_s)$ with $\mathrm{Hom}(I, I^*)$. Under this identification, the displayed derivative is symmetrization. Symmetrization is onto $\mathrm{Sym}^2(I^*)$ because two is invertible. Thus the derivative from $\mathrm{Hom}(I, X/I)$ onto $\mathrm{Sym}^2(I^*)^5$ is surjective.

Given any first order change in the five quadrics, choose δ to cancel their restrictions to I. The incidence projection is smooth and submersive at the constructed point. Its image contains a nonempty open set, which proves dominance. □

The same tangent construction can impose pairwise disjoint varying spaces I_0, I_1, I_2, I_3. Pairwise disjointness is open, and four three spaces fit in X. This remains a statement about the incidence projection in which the four spaces vary with the target tuple.

Lemma 3.11 (Oil compatible cross extension for varying spaces)**.** *Let $I_a, I_b \subseteq X$ be disjoint three spaces. Every symmetric form S on X is the quadratic form of a bilinear block C satisfying*

$$C|_{I_a \times I_b} = 0.$$

Proof. Define an alternating form A_0 on $I_a \oplus I_b$ by

$$A_0(u, v) = -S(u, v) \qquad (u \in I_a,\ v \in I_b),$$

and extend it to X. Put $C = S + A_0$. Then C vanishes on $I_a \times I_b$. Since $A_0(x, x) = 0$, the quadratic polynomial of C is the one defined by S. □

Remark 3.12 (Why the varying slice calculation is not a fixed map theorem). For a fixed realization T, the spaces $I_a = \ker \overline{T}_a$ are fixed. The diagonal image then consists of forms that vanish on these fixed spaces. The incidence projection in Theorem 3.10 instead chooses I_a after seeing the target five tuple. It can therefore be dominant even though every fixed fiber is a proper linear subspace. The cross extension in Theorem 3.11 has the same varying space qualification. These calculations may support an analysis that varies the slice with the coefficients, but they do not prove surjectivity or dominance for the fixed map Φ_T.

4 The Hilbert coefficient calculation

The homogeneous model has 50 quadrics in 31 variables. Its formal semi regular Hilbert series is

$$H_{31}(t) = \frac{(1-t^2)^{50}}{(1-t)^{31}}.$$

Expanding through degree eight gives

$$H_{31}(t) = 1 + 31t + 446t^2 + 3906t^3 + 22801t^4 + 89807t^5 \\ + 216992t^6 + 139872t^7 - 1166808t^8 + O(t^9).$$

The first negative coefficient occurs at degree eight.

If one homogenizes the affine system with one extra variable, the formal series becomes

$$H_{32}(t) = \frac{(1-t^2)^{50}}{(1-t)^{32}}.$$

Its expansion is

$$H_{32}(t) = 1 + 32t + 478t^2 + 4384t^3 + 27185t^4 + 116992t^5 \\ + 333984t^6 + 473856t^7 - 692952t^8 + O(t^9).$$

This model also selects degree eight.

Proposition 4.1 (Exact coefficient formula)**.** *For $n \in \{31, 32\}$, the coefficient of t^d in $H_n(t)$ is*

$$[t^d]H_n(t) = \sum_{j=0}^{\lfloor d/2 \rfloor} (-1)^j \binom{50}{j} \binom{n+d-2j-1}{d-2j}.$$

The two displayed expansions follow from this formula.

Proof. Use the binomial expansions

$$(1-t^2)^{50} = \sum_{j=0}^{50} (-1)^j \binom{50}{j} t^{2j}, \qquad (1-t)^{-n} = \sum_{r \geq 0} \binom{n+r-1}{r} t^r,$$

and collect the terms with $2j + r = d$. □

5 Logical status

The results in this appendix have the following scope.

1. The degree two theorem proves exact affine coefficient support on every transverse coset. It also gives the exact 19^{-4} incidence loss for the Level I direction of dimension 46.

2. The degree four theorem gives an exact fixed map criterion. The ten channel map is onto if and only if all four maps to the vinegar quotients V_a/O_a are injective.

3. After six variables are fixed, the main space has dimension 31 and each vinegar quotient has dimension 28. Full homogeneous coefficient support is therefore impossible for every fixed realization. Fixing at least nine variables removes this dimension obstruction, but one must still prove that the constrained realization locus meets the open good locus.

4. The codimension and tangent incidence calculations in Section 3.4 permit the slice intersections to vary with the target quadrics. They do not prove dominance for a fixed coefficient map.

5. Faithfully flat descent applies only after the assembled split map has been identified as the scalar extension of a base field map. It does not produce a base field slice and does not give a distributional claim for the official finite source.

6. The Hilbert expansions are exact formal calculations. The claim that a production system follows either series through degree eight remains the ordinary semi regularity assumption. The fixed map theorem does not imply it.

7. This appendix does not prove the Macaulay matrix ranks used by PXL. It does not prove that symbolic preprocessing and later variable fixing preserve the random MQ rank model. It does not provide a PXL operation count.

8. Surjectivity gives full support only when the native source has full support. Uniform output further requires a uniform native source. The official byte reduction source is not uniform on $\mathbb{F}_{19}$.

These limits separate the algebraic proof from the solver model. Any later manuscript statement must keep the same separation.